Brahim Redouane MELLAH

Senior Software Developer · C++ · Computer Vision · ML & Deep Learning · Generative AI

Montréal, QC, Canada · +1 514 239 3776 · brahim.redouane.mellah@gmail.com · ibraverse.ca · linkedin.com/in/brahim-redouane-mellah

SUMMARY

Senior software developer at Zebra Technologies, with seven years of production C++ at Matrox Imaging and Zebra. I take features from research to production: API design, testing, CI/CD, documentation, and support for the UI and documentation teams. I started with 2D and 3D image processing algorithms, then added machine learning and deep learning, and brought the library to smart cameras with little memory. Today I also build generative AI: agentic workflows, RAG, and the evaluations that decide when an agent is ready to ship. Rated "exceeds expectations" in four performance reviews. Background in robotics and control theory.

EXPERIENCE

Zebra Technologies — Montréal, QC

Senior Software Developer, Generative AI — Aug 2026 — Present

- Implement agentic workflows end to end with Google ADK and LlamaIndex: the agent loop, the tools each agent can call, its memory, and the context it receives, sized for small models.
- Build retrieval-augmented generation (RAG) with grounded answers, and run small models locally.
- Evaluate agents: build the evaluation sets and metrics that decide whether an agent is good enough to ship, the same discipline I used for deep learning models.

Senior Software Developer, Aurora Imaging Library — Jul 2024 — Aug 2026

- Implemented decision trees, random forests and k-means in the library's C++, multithreaded, thread-safe, and with early stopping.
- Benchmarked deep learning models for object detection, OCR and text detection on CPU and GPU, tuned their hyperparameters, and designed the metrics that decide whether a model ships; took the accepted ones to production.
- Brought the library to Zebra's Aurora Focus smart cameras within tight memory and CPU limits, and fixed concurrency bugs from the user interface to the backend.
- Owned vision algorithms (Edge Finder, Geometric Matcher, Calibration) from feature to release: API design, customer debugging, documentation and examples, and backward compatibility agreed with the UI, QA and documentation teams.

Software Developer II, Image Processing — Jun 2022 — Jun 2024

- Took the Advanced Geometric Matcher (AGM), a new pattern-matching tool, from research and prototype to production: designed its public C++ API with the team and shipped it.
- Added random forests and SVMs to purely geometric tools, so they could classify patterns in cluttered scenes.
- Fixed numerical bugs and API regressions, and added unit, integration and randomized tests to the regression suites.
- Mentored interns and junior developers in C++, multithreading and code review.

Matrox Imaging (acquired by Zebra in 2022) — Dorval, QC

Software Developer, Image Processing — Jun 2019 — Jun 2022

- Built and debugged the calibration, measurement and metrology tools; maintained Model Finder, Edge Finder and Bead.
- Rewrote older geometric solvers with non-linear optimization (Levenberg–Marquardt) for sub-pixel accuracy and stable results.
- Fixed memory leaks and dangling references in a decades-old C++ codebase, and built regression tests that protect results thousands of customer applications rely on.

Software Development Intern — Jan 2019 — May 2019

- Built a random forest classifier into the Matrox Imaging Library, with multithreaded training and warm start, plus synthetic datasets and automated benchmarks for precision and speed.

SKILLS

Languages	C++ (C++17/20), Python, C, TypeScript, MATLAB
Software engineering	API and SDK design, multithreading and concurrency, memory optimization, unit and integration testing, CI/CD, smart cameras, Windows and Linux
Vision & maths	2D and 3D image processing, calibration, metrology, edge detection, pattern matching; non-linear optimization (Levenberg–Marquardt), linear algebra
ML & deep learning	Decision trees, random forests, SVMs and k-means in C++; model evaluation and benchmarking on CPU and GPU; hyperparameter tuning; object detection, OCR; scikit-learn
Generative AI	Agentic workflows, RAG, agent evaluation, small local models; Google ADK, LlamaIndex, Gemini, MCP
Tools	Git, GitHub, SVN, Jira, Confluence, Visual Studio, VS Code, Docker, Google Cloud, Firebase

EDUCATION

Master of Engineering (M.Eng.), Electrical Engineering — Control Systems · Polytechnique Montréal 2017 — 2019
GPA 3.87/4 · Al Ghurair Foundation scholarship, 1 of ~100 recipients from 15,000+ applicants · ended with the Matrox internship. Hands-on projects: control of a 6-axis FANUC robot, real-time control on QNX in C++, SIFT panorama stitching, multi-robot formation control on the Georgia Tech Robotarium.

Engineering degree (Diplôme d'ingénieur d'État), Electrical Engineering — Control Systems · École Nationale Polytechnique, Algiers 2012 — 2017
GPA 17.5/20 · ranked 5th of 1,400 in the national entrance exam (top 1%) · final-year project: adaptive cruise control for autonomous vehicles, at CDTA.

CERTIFICATES & LANGUAGES

DeepLearning.AI: Neural Networks and Deep Learning; Convolutional Neural Networks (2023) · University of Washington: Machine Learning: Classification (2023)
French (native) · English (professional) · Arabic (native)

SELECTED PROJECTS

MemoryTracer — where a Windows process's memory goes ibraverse.ca/projects/memorytracer

- A C++ library that records where a Windows process's virtual memory goes over time, to find leaks and fragmentation. Reads the Windows memory APIs directly (VirtualQuery, Tool Help) and exports CSV by memory state and type. Built after a production out-of-memory case.

Farkad — a health tracker you talk to ibraverse.ca/projects/farkad

- One spoken sentence becomes structured, editable health entries: a pipeline of AI agents routes it, extracts each topic in parallel and checks it against a schema; an answer that fails goes to a stronger model. Gemini, Claude and GPT behind one interface; Python, Flutter, Firebase.
- Scored on a gold set in English, French, Arabic, Spanish and mixed languages: 100% routing and field accuracy on 15 sentences.

Domia — an agent that tests applications like a person ibraverse.ca/projects/domia

- An AI agent that tests web and desktop apps the way a person does: it reads the accessibility tree, acts, and checks what changed. A deterministic pass settles 20 findings in about 200 ms before any model is called. TypeScript, Playwright, MCP.

scikit-learn: an infinite loop in K-Means ibraverse.ca/tech/scikit-learn-kmeans-infinite-loop

- Found an input that makes K-Means loop forever, traced it to how empty clusters are handled, and reported it (issue #28055). The fix shipped in scikit-learn 1.4.1.